

Lab 18: Unsupervised Learning

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18.1 NBA Role Replacement Search

18.1.1 Overview

Lecture 18 used PCA and clustering to compare NBA players by offensive role. In this lab, you will use the same idea for three roster questions from the 2022 offseason:

- the **Detroit Pistons** replacing **Jerami Grant** (2022);
- the **San Antonio Spurs**, replacing **Dejounte Murray** (2022);
- the **Utah Jazz**, replacing **Donovan Mitchell** (2022).

The goal is not to rank the best available players. The goal is to build role-comparison lists: which recent player-seasons had offensive usage patterns most similar to the player each team lost? Your final write-up should give recommendations for all three teams and explain what this data can, and cannot, tell us.

18.1.2 The Data

The dataset contains NBA player-seasons from 2012–2022. Each row is one player in one season. The file `18_nba-clusters.csv` includes identifying columns such as `PLAYER_NAME`, `szn`, and `player_szn`, along with play-type frequency features describing how each player used his shot attempts:

- spot-ups
- pick-and-roll ball-handler possessions
- pick-and-roll roll-man possessions
- isolations
- post-ups
- cuts
- offensive rebounds
- off-screen or handoff actions

Each value is a frequency, so the features describe offensive role rather than scoring volume or efficiency. Since they say little about salary, contract status, defense, health, availability, age, or shooting efficiency, they are useful for role comparison but incomplete for roster decisions.

18.1.3 Finding Replacement Candidates

1. Create at least two visualizations that help you understand the player-season data and the play-type features.
2. Choose the numerical features that define offensive role. Standardize them by subtracting each feature's mean and dividing by its standard deviation. Explain what kind of basketball similarity your feature choices encode.
3. Run principal component analysis on the standardized features. Make a scree plot and a plot of player-seasons on the first two principal components. Highlight the three target players. Interpret what the first two components appear to measure.
4. Cluster the player-seasons using k-means. Choose a value of k , then justify that choice using an elbow plot, silhouette scores, interpretability, or some combination of these.
5. Interpret the clusters using visualizations and/or the players assigned to each cluster. Remember that the algorithm only returns similarity groups. The basketball labels come from your interpretation.
6. Using Euclidean distance on the standardized features, find the 10 closest recent player-seasons to each target. Make one small table for each team.
7. For each team, recommend 2–3 players (based on their 2021 or 2022 seasons) you would investigate further as role replacements. Explain why they are plausible comparisons, and describe what information a real front office would still need before acting on your recommendation.